

Finite Difference Methods for Option Pricing

— *Lecture Notes: Three Sessions of 50 Minutes* —

Topics Covered:

Forward / Backward / Central Differences · Explicit FDM · Implicit
FDM

Crank–Nicolson · Leapfrog · von Neumann Stability Analysis

Consistency · Stability · Convergence (Lax–Wendroff)

Connection to Binomial / Trinomial Trees

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Contents

1	Motivation: Why Finite Difference Methods?	3
1.1	When Closed-Form Solutions Fail	3
1.2	The Black–Scholes PDE	3
1.3	Logarithmic Transformation	3
2	Finite Difference Approximations	4
2.1	Taylor Expansion Foundation	4
2.2	Forward Difference (First-Order Accurate)	4
2.3	Backward Difference (First-Order Accurate)	4
2.4	Central Difference (Second-Order Accurate)	4
2.5	Second Derivative (Central, Second-Order Accurate)	4
2.6	Grid Setup	5
3	Explicit Finite Difference Method	5
3.1	Discretisation	5
3.2	Jump-Process Interpretation	6
3.3	Stability Conditions for the Explicit Scheme	6
4	Implicit Finite Difference Method	6
4.1	Discretisation	6
4.2	Tridiagonal System	7
4.3	Gaussian Elimination for Tridiagonal Systems	7
4.4	Solving the Recurrence for b_i^*	8
4.5	Jump-Process Interpretation of the Implicit Scheme	8
4.6	Unconditional Stability of the Implicit Scheme	8
5	Crank–Nicolson Method	8
5.1	Motivation: Second-Order Time Accuracy	8
5.2	Derivation	9
5.3	The θ -Method (General Framework)	9
5.4	Accuracy Comparison via the Model Problem	9
6	Leapfrog Method and Its Instability	10
6.1	Leapfrog Discretisation	10
6.2	Characteristic Equation	10
6.3	Proof of Unconditional Instability	10
7	Consistency, Stability, and Convergence	10
7.1	Three Fundamental Concepts	10
7.2	Lax–Wendroff Theorem	11
7.3	Consistency of Explicit, Implicit, Crank–Nicolson	11
8	von Neumann Fourier Stability Analysis	11
8.1	Principle	11
8.2	Explicit Scheme (Heat Equation)	12
8.3	Explicit Scheme for the BS Equation	12
8.4	Implicit Scheme (Heat Equation)	12

8.5	Crank–Nicolson Scheme (Heat Equation)	13
8.6	Leapfrog Scheme: von Neumann Analysis	13
9	Binomial / Trinomial Trees and FDM	13
9.1	CRR Binomial Tree (Review)	13
9.2	Trinomial Tree	13
9.3	Convergence of Tree Models to GBM	14
10	Summary and Comparison	14
11	Homework Problems	15
12	Black–Scholes PDE: Derivation, Boundary Conditions, and Finite Difference Schemes	16
12.1	Putting It All Together: How Previous Sections Feed into Black–Scholes	16
12.2	Derivation of the Black–Scholes PDE	17
12.3	Time Reversal and Grid Setup	18
12.3.1	Reversing Time: $\tau = T - t$	18
12.3.2	Computational Grid	18
12.3.3	Initial Condition (Payoff at $\tau = 0$)	18
12.3.4	Boundary Conditions	18
12.4	Explicit FTCS Scheme for Black–Scholes	18
12.5	Fully Implicit Scheme for Black–Scholes	19
12.5.1	Tridiagonal Matrix System	19
12.6	Crank–Nicolson Scheme for Black–Scholes	20
12.7	American Options: Early-Exercise Enforcement	20
12.8	Numerical Computation of Greeks	21
A	Appendix A: Derivation of the Implicit Scheme Coefficients	21
B	Appendix B: Full Proof of Stability for the θ-Method	22
C	Appendix C: Brennan–Schwartz (1978) – Key Results	23

1 Motivation: Why Finite Difference Methods?

1.1 When Closed-Form Solutions Fail

The Black–Scholes formula gives an analytical price for European options, but in many realistic situations no closed form is available:

- **American options** – early exercise feature
- **Barrier / lookback options** – path-dependent payoffs
- **Options on dividend-paying stocks** (Schwartz 1977)
- **Stochastic-volatility models** (Heston, SABR, ...)

Three broad numerical approaches exist:

1. **Tree models** (Binomial/Trinomial) – discretise the stock-price process.
2. **Monte Carlo simulation** – simulate paths.
3. **Finite Difference Methods (FDM)** – discretise the PDE directly.

Core Idea of FDM

Replace every continuous derivative in the PDE by an algebraic approximation on a finite grid, then solve the resulting system of equations by backward induction.

1.2 The Black–Scholes PDE

For a European option with value $V(S, t)$:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0, \quad (1)$$

with terminal condition $V(S, T) = \max(S - K, 0)$ (call) or $\max(K - S, 0)$ (put).

Boundary conditions (call):

$$V(0, t) = 0, \quad V(S, t) \approx S - Ke^{-r(T-t)} \text{ as } S \rightarrow \infty.$$

1.3 Logarithmic Transformation

The coefficients S^2 and S in (1) vary with the state variable, complicating finite-difference stencils. Setting

$$y = \ln S, \quad W(y, t) = V(S, t)$$

and applying the chain rule:

$$\frac{\partial V}{\partial S} = \frac{1}{S} W_y = e^{-y} W_y, \quad (2)$$

$$\frac{\partial^2 V}{\partial S^2} = \frac{\partial}{\partial S} \left(\frac{W_y}{S} \right) = -\frac{W_y}{S^2} + \frac{W_{yy}}{S^2} = e^{-2y} (W_{yy} - W_y), \quad (3)$$

$$\frac{\partial V}{\partial t} = W_t. \quad (4)$$

Substituting into (1) (recall $S = e^y$):

$$\frac{1}{2}\sigma^2 e^{2y} \cdot e^{-2y}(W_{yy} - W_y) + r e^y \cdot e^{-y}W_y + W_t - rW = 0.$$

Log-Transformed BS Equation

$$\frac{1}{2}\sigma^2 W_{yy} + \left(r - \frac{1}{2}\sigma^2\right) W_y + W_t - rW = 0. \quad (5)$$

All coefficients are now *constants*, independent of S .

2 Finite Difference Approximations

2.1 Taylor Expansion Foundation

Let f be sufficiently smooth. Then:

$$f(x_0 + h) = f(x_0) + f'(x_0)h + \frac{f''(x_0)}{2}h^2 + \frac{f'''(x_0)}{6}h^3 + \dots, \quad (6)$$

$$f(x_0 - h) = f(x_0) - f'(x_0)h + \frac{f''(x_0)}{2}h^2 - \frac{f'''(x_0)}{6}h^3 + \dots. \quad (7)$$

2.2 Forward Difference (First-Order Accurate)

From (6):

$$f'(x_0) \approx \frac{f(x_0 + h) - f(x_0)}{h}, \quad \text{error} = O(h). \quad (8)$$

Uses values at x_0 (known) and $x_0 + h$ (future); truncation error $O(h)$.

2.3 Backward Difference (First-Order Accurate)

From (7):

$$f'(x_0) \approx \frac{f(x_0) - f(x_0 - h)}{h}, \quad \text{error} = O(h). \quad (9)$$

2.4 Central Difference (Second-Order Accurate)

Subtracting (7) from (6):

$$f(x_0 + h) - f(x_0 - h) = 2hf'(x_0) + \frac{h^3}{3}f'''(x_0) + \dots$$

$$f'(x_0) \approx \frac{f(x_0 + h) - f(x_0 - h)}{2h}, \quad \text{error} = O(h^2). \quad (10)$$

2.5 Second Derivative (Central, Second-Order Accurate)

Adding (6) and (7):

$$f(x_0 + h) + f(x_0 - h) = 2f(x_0) + h^2f''(x_0) + O(h^4)$$

$$f''(x_0) \approx \frac{f(x_0 + h) - 2f(x_0) + f(x_0 - h)}{h^2}, \quad \text{error} = O(h^2). \quad (11)$$

2.6 Grid Setup

For equation (5), define:

- h : spatial step (log-price direction).
- k : time step.
- $W_{i,j} \approx W(ih, jk)$: approximate value at grid node (ih, jk) .

Boundary condition: at maturity $j = J$, $W_{i,J} = \max(e^{ih} - K, 0)$ (call).

Time direction: backward induction from $j = J$ down to $j = 0$.

3 Explicit Finite Difference Method

3.1 Discretisation

In the explicit scheme the spatial derivatives are evaluated at the *known* time level $j + 1$, while only $W_{i,j}$ is unknown:

$$W_y \approx \frac{W_{i+1,j+1} - W_{i-1,j+1}}{2h} \quad (\text{central difference at } j+1), \quad (12)$$

$$W_{yy} \approx \frac{W_{i+1,j+1} - 2W_{i,j+1} + W_{i-1,j+1}}{h^2} \quad (\text{central difference at } j+1), \quad (13)$$

$$W_t \approx \frac{W_{i,j+1} - W_{i,j}}{k} \quad (\text{backward in time}). \quad (14)$$

Substituting into (5) and multiplying through by k :

$$\frac{1}{2}\sigma^2 \left(\frac{W_{i+1,j+1} - 2W_{i,j+1} + W_{i-1,j+1}}{h^2} \right) + \left(r - \frac{1}{2}\sigma^2 \right) \left(\frac{W_{i+1,j+1} - W_{i-1,j+1}}{2h} \right) + \frac{W_{i,j+1} - W_{i,j}}{k} - rW_{i,j} = 0.$$

Collecting $W_{i,j}$ on the left:

Explicit Scheme – Brennan and Schwartz 1978

$$(1 + rk) W_{i,j} = a W_{i-1,j+1} + b W_{i,j+1} + c W_{i+1,j+1}, \quad (15)$$

where

$$\begin{aligned} a &= \left[\frac{1}{2} \left(\frac{\sigma}{h} \right)^2 - \frac{1}{2h} \left(r - \frac{1}{2}\sigma^2 \right) \right] k, \\ b &= \left[1 - \left(\frac{\sigma}{h} \right)^2 k \right], \\ c &= \left[\frac{1}{2} \left(\frac{\sigma}{h} \right)^2 + \frac{1}{2h} \left(r - \frac{1}{2}\sigma^2 \right) \right] k. \end{aligned} \quad (16)$$

Verification: $a + b + c = 1$. Direct algebra using (16):

$$a + b + c = \left[\frac{\sigma^2 k}{h^2} - \frac{k}{2h} \left(r - \frac{1}{2}\sigma^2 \right) \right] + \left[1 - \frac{\sigma^2 k}{h^2} \right] + \left[\frac{\sigma^2 k}{h^2} + \frac{k}{2h} \left(r - \frac{1}{2}\sigma^2 \right) \right] \cdot \frac{1}{2} + \dots$$

Careful grouping yields $a + b + c = 1$. □ □

3.2 Jump-Process Interpretation

Dividing (15) by $1 + rk$:

$$W_{i,j} = \frac{1}{1 + rk} (p^- W_{i-1,j+1} + p W_{i,j+1} + p^+ W_{i+1,j+1}), \quad (17)$$

with $p^- = a$, $p = b$, $p^+ = c$.

This is exactly the *risk-neutral expectation formula* for a three-point jump process (Cox & Ross 1976): the option today is the discounted expected value under the measure (p^-, p, p^+) .

Explicit FDM $\leftrightarrow$ Trinomial Tree

Equation (17) is numerically identical to a *trinomial tree* with jumps $\{-h, 0, +h\}$ and risk-neutral probabilities (p^-, p, p^+) under log-price $y = \ln S$. This is the central synthesis result of Brennan & Schwartz (1978).

3.3 Stability Conditions for the Explicit Scheme

For p^-, p, p^+ to be interpretable as probabilities, all must be non-negative.

Condition on $b \geq 0$:

$$1 - \left(\frac{\sigma}{h}\right)^2 k \geq 0 \implies k \leq \frac{h^2}{\sigma^2}.$$

Condition on $a \geq 0$ (equivalently $c \geq 0$):

$$\frac{1}{2} \left(\frac{\sigma}{h}\right)^2 \geq \frac{1}{2h} \left|r - \frac{1}{2}\sigma^2\right| \implies h \leq \frac{\sigma^2}{\left|r - \frac{1}{2}\sigma^2\right|}.$$

Explicit Stability Conditions – Brennan & Schwartz (1978)

$$h \leq \frac{\sigma^2}{\left|r - \frac{1}{2}\sigma^2\right|}, \quad k \leq \frac{\sigma^2}{\left(r - \frac{1}{2}\sigma^2\right)^2}. \quad (18)$$

Conditionally Stable

The explicit method is **conditionally stable**: if h or k violates (18), numerical errors grow exponentially and the solution diverges.

4 Implicit Finite Difference Method

4.1 Discretisation

In the implicit scheme, the spatial derivatives are evaluated at the *unknown* time level j :

$$W_y \approx \frac{W_{i+1,j} - W_{i-1,j}}{2h}, \quad (19)$$

$$W_{yy} \approx \frac{W_{i+1,j} - 2W_{i,j} + W_{i-1,j}}{h^2}, \quad (20)$$

$$W_t \approx \frac{W_{i,j+1} - W_{i,j}}{k}. \quad (21)$$

Substituting into (5) and rearranging:

Implicit Scheme

$$aW_{i-1,j} + bW_{i,j} + cW_{i+1,j} = W_{i,j+1}, \quad (22)$$

where

$$\begin{aligned} a &= -\left[\frac{1}{2} \left(\frac{\sigma}{h} \right)^2 - \frac{1}{2h} \left(r - \frac{1}{2} \sigma^2 \right) \right] k, \\ b &= 1 + \left(\frac{\sigma}{h} \right)^2 k + rk, \\ c &= -\left[\frac{1}{2} \left(\frac{\sigma}{h} \right)^2 + \frac{1}{2h} \left(r - \frac{1}{2} \sigma^2 \right) \right] k. \end{aligned} \quad (23)$$

4.2 Tridiagonal System

For each time step j , equation (22) with $i = 1, \dots, n$ gives the linear system $\mathbf{A}\mathbf{w}_j = \mathbf{f}_j$:

$$\underbrace{\begin{pmatrix} b & c & & & \\ a & b & c & & \\ & \ddots & \ddots & \ddots & \\ & & a & b & c \\ & & & a & b \end{pmatrix}}_{\mathbf{A}} \begin{pmatrix} W_{1,j} \\ \vdots \\ W_{n,j} \end{pmatrix} = \begin{pmatrix} W_{1,j+1} \\ \vdots \\ W_{n,j+1} \end{pmatrix} - \begin{pmatrix} aW_{0,j} \\ 0 \\ \vdots \\ 0 \\ cW_{n+1,j} \end{pmatrix}.$$

Since $\mathbf{A}$ is *independent of j* , its LU factorisation is performed **only once**.

4.3 Gaussian Elimination for Tridiagonal Systems

Forward elimination ($i = 2, \dots, n$):

Define updated coefficients starting from $b_1^* = b$, $c_1^* = c$, $f_1^* = f_1$:

$$b_i^* = \frac{b}{a} b_{i-1}^* - c_{i-1}^*, \quad c_i^* = \frac{c}{a} b_{i-1}^*, \quad f_i^* = \frac{f_i}{a} b_{i-1}^* - f_{i-1}^*. \quad (24)$$

The reduced system is upper-bidiagonal:

$$b_i^* W_{i,j} + c_i^* W_{i+1,j} = f_i^*, \quad i = 1, \dots, n.$$

Back substitution:

$$W_{n,j} = \frac{f_n^*}{b_n^*}, \quad W_{i,j} = \frac{f_i^* - c_i^* W_{i+1,j}}{b_i^*}, \quad i = n-1, \dots, 1. \quad (25)$$

4.4 Solving the Recurrence for b_i^*

The recurrence in (24) is a second-order linear recurrence:

$$b_i^* - \frac{b}{a}b_{i-1}^* + \frac{c}{a}b_{i-2}^* = 0.$$

Its characteristic equation is:

$$\lambda^2 - \frac{b}{a}\lambda + \frac{c}{a} = 0 \implies \lambda_{1,2} = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Define $L_i = \lambda_1^i - \lambda_2^i$. Then:

$$b_i^* = \frac{a^2}{\sqrt{b^2 - 4ac}} L_{i+1}.$$

4.5 Jump-Process Interpretation of the Implicit Scheme

Brennan & Schwartz (1978) show that as the grid becomes fine ($n, i \rightarrow \infty$), the inverse $\mathbf{A}^{-1}$ defines generalised “probabilities” p_ℓ^* such that:

$$W_{i,j} = (1 - rk) \sum_{\ell=-\infty}^{\infty} p_\ell^* W_{i+\ell, j+1}.$$

Theorem 4.1 (Brennan–Schwartz 1978). *Under condition $h^2 \leq \sigma^4/(r - \frac{1}{2}\sigma^2)^2$, all limiting probabilities $p_\ell^* \geq 0$, and*

$$\mathbb{E}[dy] = \left(r - \frac{1}{2}\sigma^2\right)k, \quad \text{Var}[dy] = \sigma^2k + \left(r - \frac{1}{2}\sigma^2\right)^2k^2. \quad (26)$$

The variance overestimates the GBM variance by a term of order k^2 , which vanishes as $k \rightarrow 0$.

4.6 Unconditional Stability of the Implicit Scheme

Implicit FDM: Unconditionally Stable

The implicit method is **unconditionally stable**: it produces bounded solutions for *any* choice of h and k . This is its main practical advantage over the explicit scheme.

5 Crank–Nicolson Method

5.1 Motivation: Second-Order Time Accuracy

Both explicit and implicit schemes are $O(k) + O(h^2)$. Crank–Nicolson achieves $O(k^2) + O(h^2)$ by averaging the spatial operators at levels j and $j + 1$.

5.2 Derivation

Satisfy the PDE at the *mid-point* in time $(i\Delta x, (j + \frac{1}{2})\Delta t)$ and approximate $[u_{xx}]^{j+1/2}$ by the arithmetic mean of its finite-difference approximations at levels j and $j + 1$:

$$\frac{U_i^{j+1} - U_i^j}{\Delta t} = \frac{1}{2} \left\{ \frac{U_{i+1}^{j+1} - 2U_i^{j+1} + U_{i-1}^{j+1}}{(\Delta x)^2} + \frac{U_{i+1}^j - 2U_i^j + U_{i-1}^j}{(\Delta x)^2} \right\}.$$

Setting $r_{CN} = \Delta t / (\Delta x)^2$ and rearranging:

$$-r_{CN}U_{i+1}^{j+1} + (2 + 2r_{CN})U_i^{j+1} - r_{CN}U_{i-1}^{j+1} = r_{CN}U_{i+1}^j + (2 - 2r_{CN})U_i^j + r_{CN}U_{i-1}^j. \quad (27)$$

5.3 The θ -Method (General Framework)

θ -Method

$$\frac{U^{j+1} - U^j}{\Delta t} = (1 - \theta) f(U^j) + \theta f(U^{j+1}), \quad 0 \leq \theta \leq 1. \quad (28)$$

θ	Method	Stability
0	Forward Euler (explicit)	Conditional
1/2	Crank–Nicolson	Unconditional
1	Backward Euler (implicit)	Unconditional

5.4 Accuracy Comparison via the Model Problem

Consider $u_t = -au$ ($a > 0$), exact solution $u(t) = u_0 e^{-at}$. The θ -method gives:

$$U^{j+1} = \lambda U^j, \quad \lambda = \frac{1 - (1 - \theta)ak}{1 + \theta ak}.$$

Compare with the exact amplification e^{-ak} via Taylor expansion (for small ak):

$$\lambda \approx 1 - ak + \theta(ak)^2 - \theta^2(ak)^3 + O((ak)^4),$$

$$e^{-ak} \approx 1 - ak + \frac{(ak)^2}{2} - \frac{(ak)^3}{6} + O((ak)^4).$$

Subtracting:

$$e^{-ak} - \lambda = \left(\frac{1}{2} - \theta\right)(ak)^2 + \left(\frac{1}{6} - \theta + \theta^2\right)(ak)^3 + O((ak)^4). \quad (29)$$

Crank–Nicolson Is Second-Order Accurate in Time

- $\theta \neq \frac{1}{2}$: leading error is $O((ak)^2) \Rightarrow$ *first-order* in time.
- $\theta = \frac{1}{2}$ (Crank–Nicolson): leading term vanishes, error is $O((ak)^3) \Rightarrow$ *second-order* in time.

6 Leapfrog Method and Its Instability

6.1 Leapfrog Discretisation

The leapfrog scheme uses a *centred* difference in *both* time and space:

$$\frac{U_i^{j+1} - U_i^{j-1}}{2\Delta t} = f(U^j). \quad (30)$$

Applied to $u_t = -au$:

$$U^{j+1} - U^{j-1} = -2ak U^j \quad (k = \Delta t).$$

6.2 Characteristic Equation

Substituting the Ansatz $U^j = \phi^j$:

$$\phi^2 + 2ak\phi - 1 = 0 \implies \phi_{1,2} = -ak \pm \sqrt{(ak)^2 + 1}.$$

General solution: $U^j = A\phi_1^j + B\phi_2^j$.

6.3 Proof of Unconditional Instability

Theorem 6.1. *When the leapfrog scheme (30) is applied to $u_t = -au$ with $a > 0$, it is **unconditionally unstable**.*

Proof. We have

$$\phi_2 = -ak - \sqrt{(ak)^2 + 1}.$$

For any $a > 0, k > 0$:

$$|\phi_2| = ak + \sqrt{(ak)^2 + 1} > 1.$$

For small ak :

$$\begin{aligned} \phi_1 &\approx 1 - ak \approx e^{-ak} \quad (\text{near true solution}), \\ \phi_2 &\approx -(1 + ak) \approx -e^{ak} \quad (|\phi_2| > 1, \text{ growing!}). \end{aligned}$$

Initial conditions $U^0 = u_0$, $U^1 \approx u_0 e^{-ak}$ produce a non-zero coefficient $B \neq 0$ for the ϕ_2 mode. Therefore $|U^j| \rightarrow \infty$ as $j \rightarrow \infty$, regardless of k . $\square$

Do NOT Use Leapfrog for Parabolic PDEs!

The leapfrog scheme is unconditionally unstable for parabolic equations (heat equation, Black–Scholes). It should never be used in option pricing.

7 Consistency, Stability, and Convergence

7.1 Three Fundamental Concepts

Definition 7.1 (Truncation Error & Consistency). Let $L(u) = 0$ be the PDE and $F(U) = 0$ the finite-difference scheme. The *truncation error* at node (ih, jk) is:

$$T_i^j(u) = F_i^j(u) - L_i^j(u).$$

The scheme is **consistent** if $T_i^j \rightarrow 0$ as $h, k \rightarrow 0$.

Definition 7.2 (Stability). The scheme is **stable** if the numerical solution does not grow unboundedly: there exists $M > 0$ (independent of h, k) such that $\|U^j\| \leq M\|U^0\|$ for all $j \geq 0$.

Definition 7.3 (Convergence). The scheme is **convergent** if $\max_{i,j} |U_i^j - u(ih, jk)| \rightarrow 0$ as $h, k \rightarrow 0$.

7.2 Lax–Wendroff Theorem

Theorem 7.1 (Lax–Wendroff Equivalence Theorem). *Suppose the continuous problem is well-posed (solution exists, is unique, and depends continuously on data), and the finite-difference scheme is consistent. Then:*

$$\text{Stability} \iff \text{Convergence}.$$

Equivalently: Consistency + Stability $\Rightarrow$ Convergence.

Practical Checklist

To prove that a scheme converges, it suffices to:

1. **Consistency:** Compute the truncation error via Taylor expansion and verify $T_i^j \rightarrow 0$ as $h, k \rightarrow 0$.
2. **Stability:** Show $|G| \leq 1$ for all ξ via von Neumann (Fourier) analysis.
3. **Convergence:** Conclude by the Lax–Wendroff theorem.

7.3 Consistency of Explicit, Implicit, Crank–Nicolson

For the heat equation $u_t = \alpha u_{xx}$, expand the explicit scheme at the exact solution u :

$$\frac{u(x, t+k) - u(x, t)}{k} = \alpha \frac{u(x+h, t+k) - 2u(x, t+k) + u(x-h, t+k)}{h^2}.$$

By Taylor expansion in k and h :

$$T = (u_t - \alpha u_{xx}) + \frac{k}{2} u_{tt} - \frac{\alpha h^2}{12} u_{xxx} + O(k^2, h^4).$$

Since the exact solution satisfies the PDE, the bracketed term vanishes and $T = O(k) + O(h^2) \rightarrow 0$: *the explicit scheme is consistent.*

Similar analysis for Crank–Nicolson yields $T = O(k^2) + O(h^2)$.

8 von Neumann Fourier Stability Analysis

8.1 Principle

Assume the error decomposes into Fourier modes:

$$\varepsilon_i^j = G^j e^{i\xi x_i}, \quad i = \sqrt{-1}, \quad \xi \in \mathbb{R}.$$

$G = G(\xi)$ is the *amplification factor*. The scheme is stable (in the ℓ^2 sense) if and only if:

$$|G(\xi)| \leq 1 \quad \forall \xi.$$

8.2 Explicit Scheme (Heat Equation)

For $u_t = \alpha u_{xx}$, the explicit scheme gives:

$$U_i^{j+1} = \lambda U_{i+1}^j + (1 - 2\lambda)U_i^j + \lambda U_{i-1}^j, \quad \lambda = \frac{\alpha k}{h^2}.$$

Substituting $\varepsilon_i^j = G^j e^{i\xi h}$:

$$G = \lambda e^{i\xi h} + (1 - 2\lambda) + \lambda e^{-i\xi h} = (1 - 2\lambda) + 2\lambda \cos(\xi h) = 1 - 4\lambda \sin^2\left(\frac{\xi h}{2}\right).$$

For $|G| \leq 1$:

$$-1 \leq 1 - 4\lambda \sin^2\left(\frac{\xi h}{2}\right) \leq 1.$$

The right inequality is trivially satisfied ($\lambda \geq 0$). The left gives $4\lambda \leq 2$, hence:

CFL Condition for Explicit Heat Equation

$$\lambda = \frac{\alpha k}{h^2} \leq \frac{1}{2} \iff k \leq \frac{h^2}{2\alpha}. \quad (31)$$

8.3 Explicit Scheme for the BS Equation

For scheme (15), define:

$$\alpha_0 = \frac{1}{2} \left(\frac{\sigma}{h} \right)^2 k, \quad \beta_0 = \frac{k}{2h} \left(r - \frac{1}{2} \sigma^2 \right),$$

so $a = \alpha_0 - \beta_0$, $b = 1 - 2\alpha_0$, $c = \alpha_0 + \beta_0$.

Substituting $\varepsilon_i^j = G^j e^{i\xi h}$:

$$\begin{aligned} G &= a e^{-i\xi h} + b + c e^{i\xi h} \\ &= 2\alpha_0 \cos(\xi h) + 2i\beta_0 \sin(\xi h) + (1 - 2\alpha_0). \end{aligned}$$

$$|G|^2 = (1 - 2\alpha_0 + 2\alpha_0 \cos \theta)^2 + 4\beta_0^2 \sin^2 \theta, \quad \theta = \xi h.$$

In the worst case ($\theta = \pi/2$, $\sin \theta = 1$):

$$|G|^2 = (1 - 2\alpha_0)^2 + 4\beta_0^2 \leq 1.$$

One can show this is equivalent to conditions (18).

8.4 Implicit Scheme (Heat Equation)

The implicit scheme gives:

$$U_i^j = (1 + 2\lambda)^{-1} (\lambda U_{i+1}^{j+1} + U_i^{j+1} + \lambda U_{i-1}^{j+1}) \implies G = \frac{1}{1 + 4\lambda \sin^2(\xi h/2)}.$$

Since the denominator ≥ 1 , we have $|G| \leq 1$ for all $\lambda > 0$: *unconditionally stable*.

8.5 Crank–Nicolson Scheme (Heat Equation)

From (27):

$$G = \frac{1 - 2r_{CN} \sin^2(\xi h/2)}{1 + 2r_{CN} \sin^2(\xi h/2)}. \quad (32)$$

Since both numerator and denominator have the form $1 \pm \delta$ ($\delta \geq 0$), we get $|G| \leq 1$ for all $r_{CN} > 0$: *Crank–Nicolson is unconditionally stable*.

8.6 Leapfrog Scheme: von Neumann Analysis

For the leapfrog PDE $u_t = f(u)$ applied to the heat equation, the amplification factor satisfies a *second-order* equation in G :

$$(2r + 1)G^2 - 4r \cos(\xi h) G + (2r - 1) = 0.$$

The product of roots equals $(2r - 1)/(2r + 1) < 1$ for $r > 0$, but the analysis of individual roots shows $|\Lambda_2| > 1$ when $r > \frac{1}{2}$ (i.e. $4r \sin^2(\xi h/2) > 2$). Even for $r < \frac{1}{2}$ the “parasitic” root $|\Lambda_2| = 1$ grows in time due to round-off, confirming unconditional instability for parabolic equations.

9 Binomial / Trinomial Trees and FDM

9.1 CRR Binomial Tree (Review)

In the Cox–Ross–Rubinstein (CRR) model with time step Δt :

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad p = \frac{e^{r\Delta t} - d}{u - d}.$$

Convergence conditions (first- and second-order moment matching of $\ln S_{t+\Delta t} | \mathcal{F}_t$):

First-order moment. $p \ln u + (1 - p) \ln d = (r - \frac{1}{2}\sigma^2)\Delta t$ (to order Δt). □ □

Second-order moment. $p(\ln u)^2 + (1 - p)(\ln d)^2 - [p \ln u + (1 - p) \ln d]^2 = \sigma^2 \Delta t$ (to order Δt). □ □

9.2 Trinomial Tree

In a trinomial tree, at each step the log-price jumps by $+h$, 0 , or $-h$ with risk-neutral probabilities p^+, p, p^- :

$$p^- + p + p^+ = 1, \quad h(p^+ - p^-) = (r - \frac{1}{2}\sigma^2)k, \quad h^2(p^+ + p^-) = \sigma^2 k + O(k^2).$$

Comparing with the explicit FDM coefficients a, b, c from (16):

Exact Equivalence

$$p^- = a, \quad p = b, \quad p^+ = c.$$

The explicit FDM on log-price $y = \ln S$ is *exactly* a trinomial tree. The stability conditions (18) are precisely the non-negativity conditions for the jump

probabilities.

9.3 Convergence of Tree Models to GBM

Under $\Delta t \rightarrow 0$ with $n\Delta t = T$ fixed, we verify both moment conditions hold automatically for the explicit FDM coefficients.

Mean (first moment):

$$\mathbb{E}[\Delta y] = h(c - a) = h \cdot \frac{k}{h} \left(r - \frac{1}{2}\sigma^2 \right) = \left(r - \frac{1}{2}\sigma^2 \right) k. \checkmark$$

Variance (second moment):

$$\text{Var}[\Delta y] \approx h^2(a + c) = h^2 \cdot \frac{\sigma^2 k}{h^2} = \sigma^2 k. \checkmark$$

By the central limit theorem, the trinomial tree converges in distribution to GBM as $\Delta t \rightarrow 0$.

10 Summary and Comparison

Scheme	Time acc.	Space acc.	Stability	Tree analogue	Cost/step
Explicit FDM	$O(k)$	$O(h^2)$	Conditional	Trinomial	$O(n)$
Implicit FDM	$O(k)$	$O(h^2)$	Unconditional	Generalised jump	$O(n)$
Crank–Nicolson	$O(k^2)$	$O(h^2)$	Unconditional	—	$O(n)$
Leapfrog	$O(k^2)$	$O(h^2)$	Unstable!	—	$O(n)$

Practical Recommendations

1. **Best accuracy & stability** $\rightarrow$ Crank–Nicolson.
2. **Simple implementation, small grid** $\rightarrow$ Explicit FDM (respect the CFL condition).
3. **Never use Leapfrog** for parabolic PDEs (BS equation, heat equation).
4. For American options: use the implicit or Crank–Nicolson scheme with the early-exercise constraint applied at each time step.

11 Homework Problems

Problem 1 – Finite Difference Approximations

(a) Derive the central-difference approximation for $f''(x_0)$, equation (11), starting from Taylor expansions, and show explicitly that the truncation error is $O(h^2)$.

(b) For $f(x) = e^x$ at $x_0 = 1$, numerically compute $f'(x_0)$ using forward, backward, and central differences with $h = 0.1, 0.01, 0.001$. Compare with the exact value e^1 and verify the convergence rates $O(h)$, $O(h)$, $O(h^2)$ respectively by computing the error ratios.

Problem 2 – Explicit FDM Stability

Given: $\sigma = 0.25$, $r = 0.05$.

(a) Compute the maximum allowable h and k for the explicit finite difference scheme using conditions (18).

(b) If $h = 0.1$ is fixed, what is the maximum time step k ?

(c) Using von Neumann analysis, write down the amplification factor $G(\xi)$ for the explicit BS scheme. For which values of α_0 and β_0 does $|G| > 1$?

(d) Implement the explicit FDM in Python/MATLAB to price a European call with $S_0 = 100$, $K = 100$, $T = 1$. Compare with the Black–Scholes formula. What happens when you violate condition (18)?

Problem 3 – Leapfrog Instability (Proof)

Apply the leapfrog scheme $\frac{U^{j+1} - U^{j-1}}{2k} = -aU^j$ to the model ODE $u_t = -au$ ($a > 0$).

(a) Derive the characteristic equation and find both roots $\phi_{1,2}$.

(b) Prove rigorously that $|\phi_2| > 1$ for all $ak > 0$.

(c) Given initial conditions $U^0 = 1$, $U^1 = e^{-ak}$, find the coefficients A, B in $U^j = A\phi_1^j + B\phi_2^j$ and show $B \neq 0$. Conclude that $|U^j| \rightarrow \infty$ as $j \rightarrow \infty$.

(d) Contrast with the true solution $u(t) \rightarrow 0$ as $t \rightarrow \infty$. What does this tell you about using leapfrog for option pricing?

Problem 4 – Crank–Nicolson Stability and Accuracy

(a) Carry out the full von Neumann analysis for the Crank–Nicolson scheme (27) applied to $u_t = u_{xx}$. Derive the amplification factor G (equation (32)) and prove $|G| \leq 1$ for all $r_{CN} > 0$.

(b) Using the θ -method (28) applied to $u_t = -au$, derive the error expression (29) in full. For which value of θ is the leading-order error minimised?

(c) Show that the Crank–Nicolson scheme for the heat equation has trunca-

tion error $O(k^2) + O(h^2)$ by Taylor expansion.

Problem 5 – Tree–FDM Connection

(a) Verify that the CRR binomial tree satisfies both moment conditions (mean and variance of $\ln S_{t+\Delta t}$) to first order in Δt using the CRR parameters $u = e^{\sigma\sqrt{\Delta t}}$, $d = e^{-\sigma\sqrt{\Delta t}}$.

(b) Show that the explicit FDM coefficients a, b, c from (16) satisfy the trinomial tree moment conditions:

$$h(c - a) = \left(r - \frac{1}{2}\sigma^2\right)k, \quad h^2(a + c) = \sigma^2k.$$

(c) Explain geometrically why the CFL condition $k \leq h^2/\sigma^2$ is equivalent to requiring $p^-, p, p^+ \geq 0$ in the trinomial tree.

(d) The JR (Jarrow–Rudd) model assumes $p = \frac{1}{2}$:

$$u = e^{(r - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}}, \quad d = e^{(r - \frac{1}{2}\sigma^2)\Delta t - \sigma\sqrt{\Delta t}}.$$

Verify this also satisfies both moment conditions and discuss its relation to the explicit FDM.

Problem 6 – Numerical Implementation (Programming)

Implement and compare the following methods for pricing a European call with $S_0 = K = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.2$:

- (a) Explicit FDM (ensure stability).
- (b) Implicit FDM (Gaussian elimination, tridiagonal solver).
- (c) Crank–Nicolson.
- (d) CRR binomial tree.

For each method, plot the error versus the number of grid points n on a log–log scale. What convergence orders do you observe? Compare with the theoretical predictions from the truncation error analysis.

Hint: For parts (a)–(c), set $y_{\min} = \ln(S_0) - 5$, $y_{\max} = \ln(S_0) + 5$, use $n = 50, 100, 200, 400$ spatial points.

12 Black–Scholes PDE: Derivation, Boundary Conditions, and Finite Difference Schemes

12.1 Putting It All Together: How Previous Sections Feed into Black–Scholes

The Black–Scholes (BS) PDE is a *parabolic* PDE with a diffusion term, an advection term, and a reaction term. Every technique developed in Sections 2–10 applies directly:

Earlier topic	Where it reappears in BS
Taylor expansions (§2)	Deriving every finite-difference stencil
Diffusion equation FTCS (§3–5)	Explicit BS update formula
Implicit / CN schemes (§4–5)	Tridiagonal system at each time step
Von Neumann stability (§6–7)	CFL-like bound on k and h
Tridiagonal solver (§4)	Thomas algorithm for implicit BS
Early-exercise constraint (§8)	Projection step for American options

Three implementation pitfalls to avoid

1. **Time direction.** The BS PDE carries a *terminal* condition $V(S, T)$, not an initial condition. Always reverse time via $\tau = T - t$ before discretising; otherwise the backward-induction logic is inverted.
2. **Domain truncation.** The natural domain is $S \in [0, \infty)$. You must choose a finite $S_{\max}$ and a boundary condition there. Too small an $S_{\max}$ introduces a large boundary error; too large wastes grid points.
3. **Payoff kink.** European call/put payoffs have a kink at $S = K$. Crank–Nicolson can produce spurious oscillations near that kink. A practical fix: run two fully implicit steps first to smooth the initial condition, then switch to Crank–Nicolson.

12.2 Derivation of the Black–Scholes PDE

Let $V(S, t)$ denote the value of an option when the underlying asset price is S at time t . Assume S follows geometric Brownian motion (GBM):

$$dS = \mu S dt + \sigma S dX, \quad dX \sim \mathcal{N}(0, 1)\sqrt{dt},$$

where σ is the (constant) volatility and μ is the drift.

Construct a **delta-hedged portfolio** $\Pi = V - \Delta S$ and choose $\Delta = \partial V / \partial S$ to eliminate the stochastic term. Applying Itô’s lemma and imposing the no-arbitrage condition $d\Pi = r\Pi dt$ yields:

Black–Scholes PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0. \quad (33)$$

The three spatial terms have distinct physical roles:

- $\frac{1}{2}\sigma^2 S^2 V_{SS}$ – **diffusion**: smooths any discontinuity in the payoff instantaneously.
- $rS V_S$ – **advection**: drifts the solution in the S -direction.
- $-rV$ – **reaction**: accounts for continuous discounting.

The drift μ does not appear: delta-hedging makes the option price independent of the investor’s view on the expected return.

12.3 Time Reversal and Grid Setup

12.3.1 Reversing Time: $\tau = T - t$

Because we know V at maturity T but want to recover it at earlier times, we introduce $\tau = T - t$, so $\partial V / \partial t = -\partial V / \partial \tau$. Equation (33) becomes

$$\frac{\partial V}{\partial \tau} = \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV, \quad (34)$$

which now has the same forward-in- τ structure as the diffusion equation – exactly the form handled by all schemes in Sections 3–5.

12.3.2 Computational Grid

Define

$$S_n = n \Delta S, \quad n = 0, 1, \dots, N_S; \quad \tau_j = j \Delta \tau, \quad j = 0, 1, \dots, N_t,$$

with $S_{\max} = N_S \Delta S$ and $T = N_t \Delta \tau$. The discrete solution is $V_{n,j} \approx V(S_n, \tau_j)$.

12.3.3 Initial Condition (Payoff at $\tau = 0$)

Contract	$V_{n,0}$
European call	$\max(S_n - K, 0)$
European put	$\max(K - S_n, 0)$
Bull spread	$\max(S_n - K_1, 0) - \max(S_n - K_2, 0)$
Straddle	$ S_n - K $

12.3.4 Boundary Conditions

Lower boundary ($S = 0$). At $S = 0$ both the diffusion and advection terms vanish. The PDE reduces to the ODE $\partial V / \partial \tau = -rV$, solved by one step of Forward Euler (cf. Section 2):

$$V_{0,j+1} = (1 - r \Delta \tau) V_{0,j}. \quad (35)$$

For a call this gives $V_{0,j} = 0$ for all j ; for a put, $V_{0,0} = K$ decays at rate r .

Upper boundary ($S = S_{\max}$). When the payoff is approximately linear for large S (e.g. a call), curvature vanishes and we impose

$$\left. \frac{\partial^2 V}{\partial S^2} \right|_{S_{\max}} = 0 \implies V_{N_S,j} = 2V_{N_S-1,j} - V_{N_S-2,j}. \quad (36)$$

Alternatively, a Dirichlet condition $V_{N_S,j} = S_{\max} - Ke^{-r\tau_j}$ can be used for a European call.

12.4 Explicit FTCS Scheme for Black–Scholes

Using the same spatial discretisation as in Sections 3–4 (central differences for V_S and V_{SS} , forward difference for V_τ), equation (34) at grid node (n, j) gives:

Explicit BS Update

$$V_{n,j+1} = \alpha_n^- V_{n-1,j} + \alpha_n^0 V_{n,j} + \alpha_n^+ V_{n+1,j}, \quad (37)$$

where, writing $S_n = n \Delta S$,

$$\begin{aligned} \alpha_n^- &= \frac{1}{2}(\sigma^2 n^2 - rn) \Delta \tau, \\ \alpha_n^0 &= 1 - (\sigma^2 n^2 + r) \Delta \tau, \\ \alpha_n^+ &= \frac{1}{2}(\sigma^2 n^2 + rn) \Delta \tau. \end{aligned}$$

This is a **direct update**: each $V_{n,j+1}$ is computed from three known values at level j – identical in structure to the diffusion-equation FTCS of Section 3. No linear system needs to be solved.

Stability. Following the same von Neumann argument as in Section 6, the scheme is **conditionally stable**. For large n (large S) the dominant constraint comes from the diffusion coefficient $\sigma^2 n^2$, requiring

$$\sigma^2 n^2 \Delta \tau \lesssim 1 \implies \Delta \tau \lesssim \frac{\Delta S^2}{\sigma^2 S_{\max}^2}.$$

This is the Black–Scholes analogue of the CFL condition from Section 4.

12.5 Fully Implicit Scheme for Black–Scholes

Evaluating all spatial terms at the *unknown* level $j + 1$:

Implicit BS Equation (one row)

$$A_n V_{n-1,j+1} + (1 + B_n) V_{n,j+1} + C_n V_{n+1,j+1} = V_{n,j}, \quad (38)$$

where

$$\begin{aligned} A_n &= \frac{1}{2}(rn - \sigma^2 n^2) \Delta \tau, \\ B_n &= (\sigma^2 n^2 + r) \Delta \tau, \\ C_n &= -\frac{1}{2}(\sigma^2 n^2 + rn) \Delta \tau. \end{aligned}$$

12.5.1 Tridiagonal Matrix System

Collecting (38) for $n = 1, \dots, N_S - 1$ produces a tridiagonal system at every time step:

$$M_{j+1}^L \mathbf{V}_{j+1} + \mathbf{Y}_{j+1}^L = \mathbf{V}_j,$$

where M_{j+1}^L is the $(N_S - 1) \times (N_S - 1)$ tridiagonal matrix with diagonal $1 + B_n$, sub-diagonal A_n , and super-diagonal C_n . The vector $\mathbf{Y}_{j+1}^L$ absorbs boundary contributions:

$$\mathbf{Y}_{j+1}^L = (A_1 V_{0,j+1}, 0, \dots, 0, C_{N_S-1} V_{N_S,j+1})^\top.$$

The solution at each step is

$$\mathbf{V}_{j+1} = (M_{j+1}^L)^{-1} (\mathbf{V}_j - \mathbf{Y}_{j+1}^L),$$

solved in $O(N_S)$ operations by the Thomas algorithm (Gaussian elimination for tridiagonals, cf. Section 4).

Stability. A von Neumann analysis identical to that in Section 5 shows $|\xi| \leq 1$ for all $\Delta\tau, \Delta S$; the implicit BS scheme is **unconditionally stable**. Time accuracy is $O(\Delta\tau)$.

12.6 Crank–Nicolson Scheme for Black–Scholes

Averaging the explicit and implicit spatial operators at levels j and $j+1$, respectively, the Crank–Nicolson discretisation of (34) is:

$$A_n^{j+1}V_{n-1,j+1} + (1 + B_n^{j+1})V_{n,j+1} + C_n^{j+1}V_{n+1,j+1} = -A_n^jV_{n-1,j} + (1 - B_n^j)V_{n,j} - C_n^jV_{n+1,j}, \quad (39)$$

where the coefficients A_n^j, B_n^j, C_n^j carry a $\frac{1}{2}$ factor relative to those in the implicit scheme. Both sides are tridiagonal, yielding:

$$\begin{aligned} M_{j+1}^L \mathbf{V}_{j+1} + \mathbf{Y}_{j+1}^L &= M_j^R \mathbf{V}_j + \mathbf{Z}_j^R, \\ \mathbf{V}_{j+1} &= (M_{j+1}^L)^{-1} (M_j^R \mathbf{V}_j + \mathbf{Z}_j^R - \mathbf{Y}_{j+1}^L). \end{aligned}$$

The boundary-condition implementation is identical to the fully implicit case (Dirichlet or $V_{SS} = 0$).

Scheme Comparison for Black–Scholes

Scheme	Stability	Time order	Per-step cost
Explicit FTCS	Conditional	$O(\Delta\tau)$	Direct update – $O(N_S)$
Fully implicit	Unconditional	$O(\Delta\tau)$	Thomas algorithm – $O(N_S)$
Crank–Nicolson	Unconditional	$O(\Delta\tau^2)$	Thomas algorithm – $O(N_S)$

All three schemes achieve $O(\Delta S^2)$ spatial accuracy through the central-difference stencil.

12.7 American Options: Early-Exercise Enforcement

An American option may be exercised at any $\tau \in [0, T]$. No-arbitrage requires the option value to satisfy the complementarity condition

$$V(S, \tau) \geq \text{Payoff}(S) \quad \forall (S, \tau).$$

Numerically, this is enforced by a **projection** (early-exercise check) at the end of every time step:

Projection Step for American Options

1. Solve the chosen linear system (implicit or CN) to obtain a *continuation value* $V_{n,j+1}^*$ for all n .
2. Apply the constraint:

$$V_{n,j+1} = \max(V_{n,j+1}^*, \text{Payoff}(S_n)).$$

This is equivalent to solving the *linear complementarity problem* (LCP) at each step. In practice, the entire modification to a European pricing loop is a single line:

```
V[1:-1] = np.maximum(V[1:-1], payoff[1:-1])
```

For **Bermudan options**, the projection is applied only on the designated exercise dates; all other steps proceed as for a European option.

12.8 Numerical Computation of Greeks

Once $\{V_{n,j}\}$ has been computed, all Greeks follow from the same central-difference formulas derived in Section 2 – applied to the *already-solved* grid:

$$\Delta \approx \frac{V_{n+1,j} - V_{n-1,j}}{2 \Delta S}, \quad (40)$$

$$\Gamma \approx \frac{V_{n+1,j} - 2V_{n,j} + V_{n-1,j}}{\Delta S^2}, \quad (41)$$

$$\Theta \approx -\frac{V_{n,j+1} - V_{n,j}}{\Delta \tau}. \quad (42)$$

These estimates inherit the $O(\Delta S^2)$ spatial accuracy of the underlying scheme. No additional solves are required.

Section Summary

What to take away from Section 11

1. The BS PDE is parabolic (diffusion + advection + reaction). Every scheme from earlier sections applies directly.
2. Always reverse time ($\tau = T - t$) to convert the terminal condition into an initial condition before discretising.
3. The explicit scheme is cheap per step but conditionally stable; both the implicit and Crank–Nicolson schemes are unconditionally stable and require one Thomas-algorithm solve per step.
4. American options add only a one-line projection at the end of each time step.
5. Greeks are recovered from the same grid by central differences – no extra computation needed.

A Appendix A: Derivation of the Implicit Scheme Coefficients

Starting from (5) with all spatial and reaction terms evaluated at the *unknown* level j (fully implicit), and the time derivative as a backward difference $(W_{i,j+1} - W_{i,j})/k$ where $j + 1$ is known and j is unknown:

$$\underbrace{\frac{1}{2}\sigma^2 \cdot \frac{W_{i+1,j} - 2W_{i,j} + W_{i-1,j}}{h^2}}_{\text{diffusion at } j} + \underbrace{\left(r - \frac{1}{2}\sigma^2\right) \cdot \frac{W_{i+1,j} - W_{i-1,j}}{2h}}_{\text{advection at } j} + \frac{W_{i,j+1} - W_{i,j}}{k} \underbrace{-rW_{i,j}}_{\text{reaction at } j} = 0.$$

Multiply through by k and collect all j -level unknowns on the left, moving $W_{i,j+1}$ to the right:

$$\left[-\frac{\sigma^2 k}{2h^2} + \frac{\left(r - \frac{1}{2}\sigma^2\right)k}{2h} \right] W_{i-1,j} + \left[1 + \frac{\sigma^2 k}{h^2} + rk \right] W_{i,j} + \left[-\frac{\sigma^2 k}{2h^2} - \frac{\left(r - \frac{1}{2}\sigma^2\right)k}{2h} \right] W_{i+1,j} = W_{i,j+1}.$$

The $W_{i,j}$ diagonal coefficient is $1 + (\sigma/h)^2 k + rk$. Absorbing the rk into the definition of b and rearranging into the form $aW_{i-1,j} + bW_{i,j} + cW_{i+1,j} = W_{i,j+1}$ yields the coefficients (23).

where, writing $\mu = r - \frac{1}{2}\sigma^2$:

$$a = -\left[\frac{1}{2}\left(\frac{\sigma}{h}\right)^2 - \frac{\mu}{2h} \right] k, \quad b = 1 + \left(\frac{\sigma}{h}\right)^2 k + rk, \quad c = -\left[\frac{1}{2}\left(\frac{\sigma}{h}\right)^2 + \frac{\mu}{2h} \right] k.$$

These are exactly the coefficients (23).

Consistency check with the explicit scheme. Compare with the explicit coefficients (16): $a_{\text{exp}} = [\frac{1}{2}(\sigma/h)^2 - \mu/(2h)]k$ and $c_{\text{exp}} = [\frac{1}{2}(\sigma/h)^2 + \mu/(2h)]k$. We have $a_{\text{imp}} = -a_{\text{exp}}$ and $c_{\text{imp}} = -c_{\text{exp}}$: moving the spatial operator from the right-hand side (explicit) to the left-hand side (implicit) negates the off-diagonal weights. The diagonal changes from $b_{\text{exp}} = 1 - (\sigma/h)^2 k$ to $b_{\text{imp}} = 1 + (\sigma/h)^2 k + rk$, which is always greater than 1, guaranteeing diagonal dominance and hence unconditional stability. Note also that the right-hand side simplifies to $W_{i,j+1}$ (not $(1 + rk)W_{i,j+1}$), because the reaction term $-rW$ is treated implicitly at level j and contributes entirely to the b coefficient on the left.

B Appendix B: Full Proof of Stability for the θ -Method

For $u_t = -au$ the θ -method gives amplification $\lambda = (1 - (1 - \theta)ak)/(1 + \theta ak)$. Stability requires $|\lambda| \leq 1$.

Right bound $\lambda \leq 1$:

$$1 - (1 - \theta)ak \leq 1 + \theta ak \iff (1 - 2\theta)ak \leq 0.$$

For $a, k > 0$ this holds when $\theta \geq \frac{1}{2}$. For $\theta < \frac{1}{2}$ it holds whenever $ak \leq 2/(1 - 2\theta)$.

Left bound $\lambda \geq -1$:

$$1 - (1 - \theta)ak \geq -(1 + \theta ak) \iff 2 \geq (1 - 2\theta)ak.$$

Stability Summary for θ -Method

- $\theta \geq \frac{1}{2}$: unconditionally stable.
- $0 \leq \theta < \frac{1}{2}$: conditionally stable, $ak \leq \frac{2}{1 - 2\theta}$.

In particular, $\theta = 0$ (explicit) requires $ak \leq 2$; for the heat equation in space this translates to the CFL condition (31).

C Appendix C: Brennan–Schwartz (1978) – Key Results

Reference: M. J. Brennan & E. S. Schwartz, “Finite Difference Methods and Jump Processes Arising in the Pricing of Contingent Claims: A Synthesis,” *J. Financial & Quantitative Analysis*, 13(3), 461–474, 1978.

Main contributions:

1. Showed that solving the BS PDE by explicit FDM is *numerically equivalent* to approximating the GBM by a trinomial jump process (Cox & Ross 1976).
2. Showed that the implicit FDM corresponds to a more general jump process; the resulting “probabilities” are non-negative under mild conditions.
3. Established stability and accuracy conditions for both schemes (equations (18) and (26)).